

Assessed Problem Sheet 2 - Week 5

This problem sheet is assessed. Submit your answers via Moodle by 5pm on Thursday 14th November 2024.

Marks per section are shown in square brackets. Be clear about what you are doing, and show your working. Use Dirac notation (or a corresponding column vector / matrix notation) for all these questions.

This problem sheet is out of 23 marks. Your mark will be rescaled to a mark out of 100 when uploaded to Moodle.

1. Ladder Operators and Expectation Values:

Consider a quantum harmonic oscillator described by the Hamiltonian:

$$\hat{H} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

where $\hat{a}_+$ and $\hat{a}_-$ are the raising and lowering operators, respectively.

(a) Show that $[\hat{a}_-, \hat{a}_+] = 1$ and explain the physical meaning of this commutation relation. [2]

(b) Calculate the uncertainty of the position operator $\Delta\hat{x}$ for the normalized state: [4]

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \frac{\sqrt{2}}{\sqrt{3}}|1\rangle$$

(c) Likewise, calculate $\Delta\hat{p}$ for the same normalized state. [4]

(d) Verify that the uncertainty principle $\Delta\hat{x}\Delta\hat{p} \geq \frac{\hbar}{2}$ holds. [1]

You might find the following useful, the position and momentum operators $\hat{x}$ and $\hat{p}$ are related to the ladder operators as:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}_+ + \hat{a}_-), \quad \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}_+ - \hat{a}_-)$$

2. Matrix Representation of Ladder Operators: In this question, we look at a different system. This system is of **two** uncoupled quantum harmonic oscillators. It has the following Hamiltonian:

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega_1^2\hat{x}^2 + \frac{\hat{p}_y^2}{2m} + \frac{1}{2}m\omega_2^2\hat{y}^2$$

For clarity:

$$a_{+\hat{x}} = (\gamma)^{\frac{1}{2}}(m\omega\hat{x} - i\hat{p}_x)$$

This two QHO system is described by the basis of states $|n_1\rangle|n_2\rangle$.

(a) Rewrite the Hamiltonian in the form $\hat{H}(a_{\pm\hat{x}}, a_{\pm\hat{y}})$. [3]

(b) The system experiences coupling given by the potential: [4]

$$V(\hat{x}, \hat{y}) = g(a_{-\hat{x}} + a_{+\hat{x}})(a_{-\hat{y}} - a_{+\hat{y}})$$

Construct a 4 by 4 matrix representation of the coupled system such that, $\hat{H} + \hat{V}$. For this question consider only the lowest four states, with zero or single quanta occupancies, i.e. when $n_1, n_2 = 0, 1$.

(c) Now considering the case when $n_2 = 1$ while $n_1 = 0, 1, 2$, give the 3 by 3 matrix representation of the coupled system for the case where the coupling potential is: [5]

$$V(\hat{x}, \hat{y}) = g_2(a_{+\hat{x}}a_{-\hat{x}}\hat{y}^2)$$